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Letter to the Editor

# The galactic acceleration scale as the Jeans rate of the vacuum

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**ABSTRACT**

*Context.* The acceleration scale $g_\dagger$ of the radial acceleration relation (RAR) lies close to $cH_0/2\pi$, a coincidence without an accepted explanation.
*Aims.* We propose a physical origin for both the value of $g_\dagger$ and the functional form of the RAR.
*Methods.* We treat the vacuum as a physical medium of density $\rho_\Lambda$ and identify $g_\dagger$ with its Jeans rate, $g_\dagger = c\sqrt{4\pi G\rho_\Lambda}/2\pi = c^2\sqrt{\Lambda/2}/2\pi = 1.056\times10^{-10}\,\mathrm{m\,s^{-2}}$. Taking the relaxed vacuum around matter to be a mode in its ground state, thermally excited by a bath at the vacuum's Jeans temperature, and requiring scale invariance far from matter yields $g = g_N/(1-e^{-\sqrt{g_N/g_\dagger}})$. We test the scale with no parameter fitted and with pass/fail rules fixed in advance.
*Results.* The scale describes 153 SPARC rotation curves, passes galaxy–galaxy lensing circular velocities and the KiDS-1000 lensing RAR of late-type and blue lenses, and is consistent with 145 discs at $z = 0.7$–$2.5$, where growth with $H(z)$ is excluded at $3.4\sigma$. It lies within the systematic uncertainty of the commonly quoted $1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$. The form $c^2\sqrt{\Lambda/3}/2\pi$ fails the three low-redshift tests.
*Conclusions.* The scale agrees with $cH_0/2\pi$ today only because the vacuum dominates the present expansion. It is predicted to stay constant for late-type lenses at $z \gtrsim 1$ and to follow the vacuum density rather than the expansion rate.

**Key words.** Gravitation – Galaxies: kinematics and dynamics – Gravitational lensing: weak – dark energy – Cosmology: theory

## 1. Introduction

Rotationally supported galaxies follow a tight radial acceleration relation (RAR) between the observed acceleration $g$ and the Newtonian acceleration $g_N$ of their baryons (McGaugh et al. 2016),

$$g = \frac{g_N}{1-e^{-\sqrt{g_N/g_\dagger}}}, \tag{1}$$

with a single scale $g_\dagger = 1.20 \pm 0.02\,(\mathrm{random}) \pm 0.24\,(\mathrm{syst}) \times 10^{-10}\,\mathrm{m\,s^{-2}}$, where the systematic term is the 20% normalisation uncertainty of the stellar mass-to-light ratio. That this scale lies close to $cH_0/2\pi$ was noted when it was first measured (Milgrom 1983). It has been related to the cosmological constant through the Unruh effect (Milgrom 1999), and to the cosmic age through a geometric factor in hyperconical modified gravity (HMG; Monjo & Banik 2024; Monjo 2025). No derivation fixes both the functional form of Eq. (1) and the value of $g_\dagger$ without an adjustable factor. This letter proposes one, and tests it.

## 2. The scale: the Jeans rate of the vacuum

A medium of density $\rho$ has a natural gravitational rate, the Jeans frequency $\omega_J = \sqrt{4\pi G\rho}$. Treating the vacuum as a physical medium of density $\rho_\Lambda = \Lambda c^2/8\pi G$ gives

$$\omega_J = c\sqrt{\Lambda/2}, \qquad g_\dagger = \frac{c\,\omega_J}{2\pi} = \frac{c\sqrt{4\pi G\rho_\Lambda}}{2\pi} = \frac{c^2\sqrt{\Lambda/2}}{2\pi}. \tag{2}$$

Here $\omega_J$ is the medium's own gravitational rate, set by its density alone. The vacuum does not collapse, and its negative pressure, which drives the expansion, does not enter this rate. With Planck 2018 parameters ($H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\Lambda = 0.685$; Planck Collaboration 2020) Eq. (2) gives $g_\dagger = 1.056\times10^{-10}\,\mathrm{m\,s^{-2}}$, within the systematic range of the measured value. Since $c^2\Lambda/2 = \frac{3}{2}\Omega_\Lambda H_0^2$, Eq. (2) equals $cH_0/2\pi$ to within 1.4% today, because $\Omega_\Lambda \simeq 2/3$. In the past $H(z)$ was larger but $\rho_\Lambda$ was not, so Eq. (2) stays constant while $cH(z)/2\pi$ grows.

The physical reading is a comparison of rates. Matter relaxes the vacuum where its own gravitational rate, $g_N/c$, exceeds the vacuum's, and the transition lies where the two are comparable.

## 3. The form: ground-state occupation and scale invariance

*Postulate.* The relaxed vacuum around a mass is a vacuum mode in its ground state. The unrelaxed vacuum is the same mode thermally excited by a bath set by the vacuum itself, at $kT = \hbar\omega_J/2\pi$, which has the form of the Gibbons–Hawking temperature (Gibbons & Hawking 1977) with $H \to \omega_J$. The mass sets the mode.

*Statistics.* For a bosonic mode with $x = E/kT$, the ground-state probability is $1-e^{-x}$ and the mean occupation is $\langle n\rangle = 1/(e^x-1)$. Identifying the relaxed fraction with the ground-state probability, and the gravitational boost with $\nu = 1/(\text{relaxed fraction})$, gives

$$\nu = \frac{1}{1-e^{-x}} = 1 + \langle n\rangle. \tag{3}$$

*Scale invariance.* Far from matter the unrelaxed vacuum is taken to be invariant under a common rescaling of time and space, the symmetry that underlies the deep-MOND regime (Milgrom 2009). Write $x = (g_N/g_\dagger)^p$. For $x \ll 1$, $\nu \simeq 1/x$ and $g \simeq g_N^{1-p}g_\dagger^p$. Under a rescaling by $\lambda$, $g_N \to g_N/\lambda^2$, while

**Table 1.** Results. Pass: $\Delta\mathrm{BIC} < 10$, $\Delta\chi^2 < 4$, $|R - R_{\mathrm{pred}}| < 2\sigma$.

| Dataset | Eq. (2) | $cH(z)/2\pi$ | $\sqrt{\Lambda/3}$ |
|---|---|---|---|
| SPARC ($\Delta$BIC)[a] | −7.8 | −7.8 | +53 |
| G–G lensing ($\Delta\chi^2$)[b] | +3.2 | +0.9 | +19.0 |
| KiDS late/blue ($\Delta\chi^2$)[c] | +1.2 | +0.2 | +13.8 |
| High-$z$ ($R = +0.33 \pm 0.20$) | $1.7\sigma$ | $3.4\sigma$, fails | $1.7\sigma$ |

**Notes.** Best single $g_\dagger$ in units of $10^{-10}\,\mathrm{m\,s^{-2}}$: [a] 1.054 (free fit); [b] $1.21^{+0.19}_{-0.17}$; [c] $1.15^{+0.19}_{-0.16}$ ($2\sigma$).

a scale-invariant acceleration must scale as $g \to g/\lambda$. This fixes $2(1-p) = 1$, so $p = \frac{1}{2}$: the mode energy is the geometric mean of the matter rate and the bath frequency. Then $g = \nu g_N$ is Eq. (1), with $g_\dagger$ from Eq. (2) and no adjustable factor. In the deep regime it gives $g = \sqrt{g_N g_\dagger}$, flat rotation curves and $v^4 = GMg_\dagger$.

## 4. Tests

All tests compare Eq. (1) with $g_\dagger$ fixed by Eq. (2) against two alternatives fixed in advance: the expansion form $cH(z)/2\pi$ and the commonly quoted vacuum form $c^2\sqrt{\Lambda/3}/2\pi = 0.863 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. Pass rules were fixed before running, and no parameter was fitted to the data being tested.

1. SPARC (Lelli et al. 2016): 153 rotation curves (3135 points), with mass-to-light ratio, distance and inclination marginalised per galaxy. Rule: $\Delta\mathrm{BIC} < 10$ relative to the same law with $g_\dagger$ free.
2. Galaxy–galaxy weak lensing: circular velocities of isolated lenses at $z \approx 0.24$–$0.39$ (Mistele et al. 2024), 20 points. Rule: $\Delta\chi^2 < 4$ from the best single $g_\dagger$.
3. KiDS-1000 lensing RAR (Brouwer et al. 2021), late-type ($n < 2.5$) and blue lenses only, 30 points at $z \approx 0.25$. Same rule.
4. High-redshift kinematics: 145 discs at $z = 0.7$–$2.5$ (Price et al. 2021; Stott et al. 2016; Wisnioski et al. 2019; Tiley et al. 2021). The statistic $R = \ln[g_\dagger(z)/g_\dagger^{\mathrm{local}}]/\ln[H(z)/H_0]$ is 0 for a constant scale and 1 for the expansion form.

The Jeans scale passes all four tests (Table 1). The expansion form fails the high-redshift test, and the $\sqrt{\Lambda/3}$ form fails the three low-redshift tests.

## 5. Discussion

*Relation to earlier work.* Milgrom (1999) related MOND inertia to the Unruh temperature of an accelerated observer in de Sitter space, using temperature differences. This gives $a_0 \propto \sqrt{\Lambda/3}$, with interpolating functions that differ from Eq. (1). In HMG the analogous scale is $a_{\gamma_0} = 2c/(\gamma_0 t)$, with $t$ the cosmic age and $\gamma_0$ a factor set by the projection geometry; matching the observed $a_0$ requires $\gamma_0 \approx 13 \pm 3$ (Monjo & Banik 2024). With HMG's $t = 1/H$ and its predicted $\Omega_\Lambda = 0.6937$ (Monjo 2018), Eq. (2) is this expression with $\gamma_0 = 4\pi/\sqrt{3\Omega_\Lambda/2} \simeq 12.3$, inside that range; the corresponding projection angle, $0.462\pi$, lies between the fitted values $0.460\pi$ and $0.468\pi$ (Monjo 2024). The present proposal differs in using occupation statistics, in deriving the observed functional form, and in fixing the factor through the vacuum's Jeans rate, $\sqrt{\Lambda/2}$, which the data prefer over $\sqrt{\Lambda/3}$. It also differs from HMG in time: a scale proportional to $1/t$ is larger at earlier epochs, whereas Eq. (2) follows $\rho_\Lambda$ and stays constant. The high-redshift test favours a constant scale, but not decisively.

*Limitations.* (i) At $z < 0.5$, Eq. (2) and $cH(z)/2\pi$ differ by about 0.05 dex, so the low-redshift tests mainly confirm $g_\dagger \simeq cH_0/2\pi$ and reject $\sqrt{\Lambda/3}$. The separation from the expansion form rests on the high-redshift test, whose four samples disagree with one another ($\chi^2 = 9.9$ for 3 degrees of freedom), and which uses standard-cosmology distances. (ii) The rejection of $\sqrt{\Lambda/3}$ depends on stellar masses, whose systematic uncertainty is comparable to the 20% difference involved. (iii) The two lensing datasets share KiDS lenses. (iv) Early-type lenses show excess lensing signal in every model, which is not addressed here; HMG attributes such departures to an environment-dependent projection angle (Monjo 2025). (v) The ground-state postulate was formulated with Eq. (1) in view; its support lies in the value and constancy of the scale, which were not fitted.

*Predictions.* (1) The scale is constant for late-type lenses at $z \gtrsim 1$, where $cH(z)/2\pi$ is 1.8 times larger at $z = 1$; Euclid- and Rubin-era weak lensing can test this. (2) The scale follows $\rho_\Lambda$: with $\Omega_m h^2 = 0.143$ fixed by the CMB, $H_0 = 73$ instead of $67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ would raise it by about 12%, against 8% for $cH_0/2\pi$, so local $H_0$ determinations and the RAR scale constrain each other. (3) If dark energy evolves, $g_\dagger \propto \sqrt{\rho_{\mathrm{DE}}(z)}$.

This proposal is developed within a broader reading of the renormalisation scale as a coordinate (One 2026), but the derivation and tests above do not depend on it.

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